IS 3920 – Individual Project on Business Solutions

Software Requirements Specification

for

EaseInn – A Comprehensive Management Solution for Small and Medium-Scale Resorts in Sri Lanka

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1/19/2026

Student's Declaration

I hereby declare that this report is my original work and has not been submitted, in whole or in part, for any degree or diploma at any university or other institution of higher learning. All information derived from the work of others, whether published or unpublished, has been properly acknowledged in the text, and a complete list of references has been provided.

1/19/2025

………………………… ……………………………………

Date Signature of Student

Supervisors' declaration

I hereby declare that I have reviewed this project and find it adequate in both scope and quality.

1. Name of Supervisor:

   Designation:

   Date:

   Signature:

   Any further comments:

2. Name of Industry Supervisor: Chaveen Ellawela

   Designation: Associate Tech Lead

   Date: 1/19/2026

   Signature: Chaveen

   Any further comments:

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1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and nonfunctional requirements for EaseInn, a resort booking and operations management platform composed of a web portal and a staff mobile application. The purpose of this document is to provide a complete, precise, and verifiable description of what the system shall do, the constraints under which it shall operate, and the quality attributes it shall meet. The SRS forms the baseline for design, implementation, testing, deployment, and academic evaluation. It also supports requirement traceability by linking project objectives to system features and measurable requirements.

EaseInn targets the operational realities of Sri Lankan small and medium-scale resorts, homestays, and cabanas. In such contexts, bookings are frequently handled through phone calls and messaging applications while internal coordination is done verbally or using spreadsheets. These approaches create errors and make it difficult to measure performance. By translating these workflows into a structured information system, EaseInn improves accuracy, accountability, and business visibility.

1.2 Document Conventions

This document follows the IEEE 830 / IEEE 29148 standards for Software Requirements Specifications. Functional requirements are described using mandatory keywords such as *shall*, indicating compulsory system behavior. Optional or recommended behaviors are described using *should*. Diagrams are represented using standard UML notation, while database models follow Entity Relationship Diagram (ERD) conventions. Headings, numbering, and formatting are kept consistent throughout the document to ensure clarity and readability.

1.3 Intended Audience and Reading Suggestions

This document is intended for a diverse audience, including academic evaluators, project supervisors, software developers, system designers, testers, and future maintainers. Evaluators and supervisors are encouraged to review Sections 1, 2, and 5 to assess scope, feasibility, and quality. Developers and designers should focus on Sections 3 and 4 for detailed functional and interface requirements. Testers and quality assurance personnel may use this document to derive test cases and validation criteria.

1.4 Product Scope

EaseInn is a cloud-based resort management system that centralizes booking management, room inventory control, payments, invoicing, staff task coordination, guest communication, and business analytics. The system is designed to be lightweight and cost-effective, reducing reliance on commission-based external platforms and minimizing training needs for non-technical users. EaseInn is not intended to be a global marketplace; instead, it focuses on internal operational excellence for SMEs. The system supports multi-property management to accommodate owners who operate more than one resort or accommodation unit.

The scope includes: (1) operational workflows such as booking creation, modification, cancellation, and calendar scheduling; (2) financial workflows such as payment requests, tracking, invoicing, and reporting; (3) staff workflows such as task assignment, checklists, and completion evidence; and (4) analytical workflows such as occupancy monitoring and revenue summaries. Integration scope is limited to payment gateways and notification services. Channel management integrations and complex enterprise integrations are excluded from the initial scope due to feasibility constraints.

1.5 Problem in brief

A significant number of small and medium-scale hospitality providers in Sri Lanka rely on manual methods such as phone calls, WhatsApp messages, notebooks, and Excel spreadsheets to manage reservations and operations. These methods are fragmented and error-prone. For example, room availability is not updated in real time, which can lead to double bookings. Guest details may be scattered across different devices or lost when staff change. Payment status is often tracked manually, increasing the risk of financial mismanagement. Staff coordination is typically verbal, which reduces accountability and makes it difficult to ensure rooms are prepared on time.

Existing global platforms assist with guest reservations but do not provide internal operational management and often charge commission fees. Enterprise property management systems provide rich features but can be expensive, complex, and not localized for Sri Lankan SMEs. Therefore, a localized and affordable digital solution is needed to centralize operations, reduce errors, and improve guest experience.

1.6 Aim and Objectives

Aim:

The primary aim of this project is to design and develop a comprehensive, centralized booking and management system that enhances operational efficiency, improves decision-making, and elevates guest satisfaction for small and medium-scale resorts in Sri Lanka.

Objectives:

- Centralize booking, room inventory, and availability management into a single platform with calendar scheduling.
- Integrate secure payment options and automated invoicing for improved financial accuracy and professionalism.
- Provide dashboards and reports for occupancy, revenue, seasonal trends, and staff/task performance.
- Implement staff task management with notifications, checklists, and evidence uploads using a mobile app.
- Support guest communications such as confirmations, reminders, and optional feedback collection.
- Enable multi-property management and scalable configuration for expanding businesses.
- Ensure system security through role-based access control, audit logs, and secure data transmission.

1.7 References

1 IEEE Std 29148-2018: Systems and Software Engineering - Requirements Engineering.
2 IS 3920 Interim Evaluation Guidelines, Department of Interdisciplinary Studies, University of Moratuwa.
3 Keycloak documentation (identity and access management).
4 PayHere/Stripe API documentation (payment integration).

2 Overall description

The EaseInn system is a cloud-based booking and management solution developed to support the operational and administrative activities of small and medium-scale resorts in Sri Lanka. The system is designed in response to the widespread reliance on manual and semi-digital methods such as phone calls, WhatsApp messages, handwritten records, and spreadsheet-based tracking, which often lead to operational inefficiencies including double bookings, loss of information, delayed communication, and financial inaccuracies.

EaseInn acts as a centralized Property Management System (PMS) that integrates all core resort operations into a single, unified platform. The system enables resort owners, managers, and staff to manage bookings, room availability, payments, invoicing, staff tasks, guest communications, and business analytics through a structured and user-friendly interface. By centralizing these functions, EaseInn improves data consistency, reduces human errors, and enhances operational transparency across the organization.

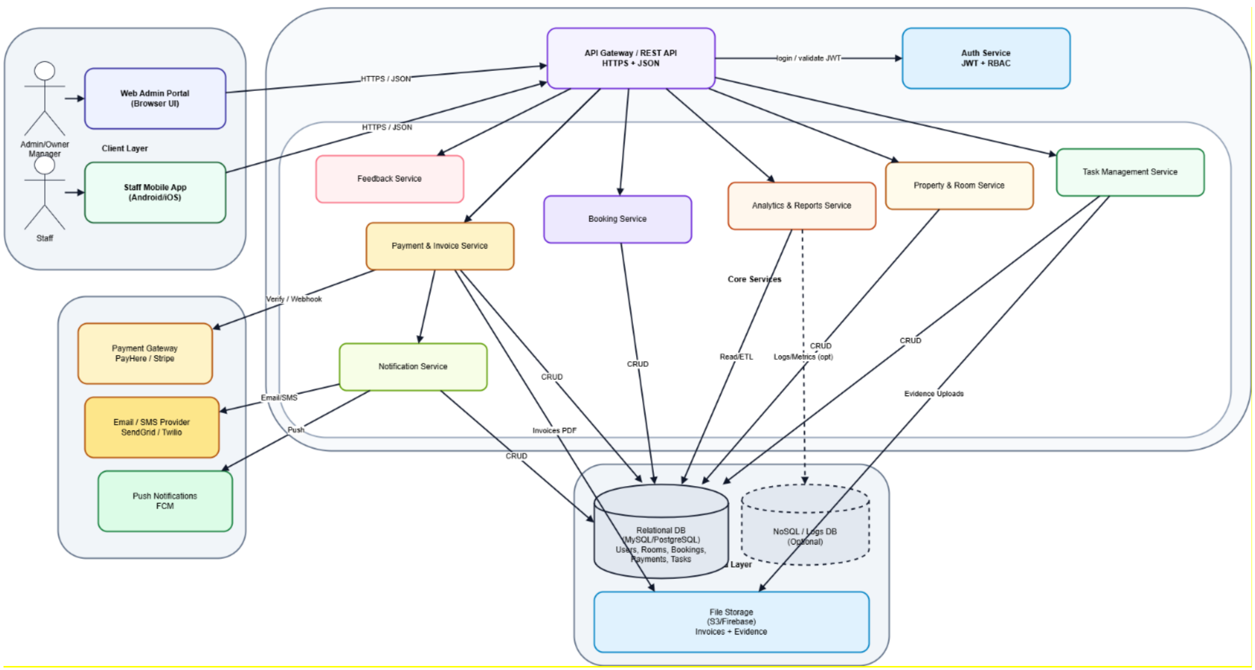
From a functional perspective, EaseInn processes inputs such as guest booking requests, payment details, staff task updates, and customer information. These inputs are handled through automated processes including reservation management, payment processing via secure gateways, task assignment and tracking, notification handling, and analytical data processing. The system then generates outputs such as real-time dashboards, invoices, payment confirmations, operational reports, notifications, and guest feedback summaries, which assist stakeholders in decision-making and service improvement.

The system follows a client–server architecture with a clear separation of concerns. A web-based portal is provided for resort owners and managers to perform administrative and managerial functions, while a mobile application is provided for staff members to receive and update task-related information. Both client interfaces communicate with the backend system through secure RESTful APIs using HTTPS and JSON data formats. Backend services handle authentication, business logic, data processing, and integrations with external systems such as payment gateways and notification providers.

EaseInn is designed as a role-based multi-user system, supporting different user classes including Admin/Owner, Manager, Staff, and Guest. Each user role has clearly defined permissions and responsibilities to ensure secure and controlled access to system features. Sensitive operations such as pricing configuration, user management, and financial reporting are restricted to authorized roles, while operational tasks are simplified for staff members with minimal technical expertise.

The system is deployed in a cloud environment, enabling scalability, availability, and cost efficiency. It supports multi-property management, allowing a single owner or organization to manage multiple resorts under one system account. External integrations such as online payment gateways, email and SMS services, and cloud storage services are used to enhance system functionality and provide a complete end-to-end solution.

In summary, the EaseInn system provides a lightweight, commission-free, and user-friendly digital platform tailored to the specific needs of Sri Lankan hospitality SMEs. By digitizing and automating critical resort operations, EaseInn improves efficiency, supports informed decision-making, enhances guest satisfaction, and contributes to the long-term sustainability and competitiveness of small and medium-scale resorts.



2.1 Product Perspective

EaseInn is a standalone, cloud-hosted information system that becomes the central operational platform within a resort organization. It supports the end-to-end flow of booking management, guest servicing, room readiness, and billing. In the wider organizational context, EaseInn reduces manual communication overhead by providing shared visibility: owners and managers can see bookings and payments in real time, while staff receive structured tasks linked to bookings and rooms.

The system follows a modular layered architecture. The presentation layer consists of a responsive web interface for owners/managers and a mobile interface for staff. The application layer exposes RESTful APIs that implement business logic including availability validation, pricing, payment reconciliation, and task workflows. The data layer stores transactional data (bookings, payments, invoices) in a relational database and stores flexible/unstructured content (audit logs, feedback, attachments) in a NoSQL store. External services include payment gateways and email/SMS notification providers. This architecture supports maintainability and scalability while remaining feasible for a one-year project.

Operationally, EaseInn supports three organizational levels. At the strategic level, dashboards and reports help owners identify trends and improve pricing decisions. At the operational level, the system manages daily bookings, check-ins, and payments. At the support level, staff tasks and notifications ensure rooms are prepared on time and guest communication remains consistent.

- **EaseInn replaces the need for:**
    - Phone-call–based booking logs
    - WhatsApp message tracking
    - Excel sheets for reservations and payments
    - Manual staff task coordination
- **The software integrates into the organization at three operational levels:**
    - Strategic level: Provides analytics and AI-driven insights to support managerial decision-making.
    - Operational level: Manages day-to-day activities such as bookings, payments, and staff tasks.
    - Support level: Enhances communication, notifications, and record keeping.

Key Product Perspectives on Functionality and Features

- **Business Perspective**
    - Acts as a centralized Property Management System (PMS) tailored for Sri Lankan SMEs.
    - Eliminates reliance on third-party commission-based platforms.
    - Improves revenue visibility and operational efficiency.
- **User Perspective**
    - Web portal designed for owners and managers with minimal IT expertise.
    - Mobile application optimized for staff with simple task-based interactions.
    - Intuitive interfaces reduce training and onboarding time.
- **Technical Perspective**
    - Follows a modular, service-oriented architecture.
    - Frontend and mobile clients communicate with backend services via RESTful APIs.
    - Authentication is handled through a centralized identity management system.
    - Supports integration with third-party services such as payment gateways and notification systems.
- **Scalability Perspective**
    - Supports multi-property management, allowing businesses to expand without changing systems.
    - Cloud-based deployment ensures scalability and availability.
- **Security Perspective**
    - Implements role-based access control to protect sensitive data.
    - Maintains audit trails for accountability and traceability.

2.2 Product Functions

- Provides a secure user authentication and authorization mechanism with role-based access control for owners, managers, and staff.
- Enables centralized booking management, allowing users to create, modify, cancel, and view reservations using a real-time calendar-based interface.
- Supports room and property management, including adding rooms, defining room types, pricing, meal plans, seasonal rates, and tracking room availability and maintenance status.
- Facilitates online payment processing, enabling the generation of secure payment links, tracking of advance and full payments, and attachment of payment references.
- Automatically generates professional invoices and receipts, which can be sent to guests via email or SMS.
- Provides a dashboard and reporting module that displays real-time metrics such as occupancy rates, revenue summaries, booking trends, and task statuses.
- Includes AI-powered analytics and insights, such as demand forecasting, seasonal trend analysis, and pricing recommendations based on historical data.
- Enables staff and task management, allowing managers to create, assign, prioritize, and track tasks linked to specific rooms or bookings.
- Supports subtasks, checklists, and task dependencies, ensuring operational activities such as cleaning and maintenance are completed correctly.
- Offers a mobile application for staff, allowing them to receive task notifications, update task statuses in real time, and upload images or notes.
- Provides offline functionality for the staff mobile application, enabling task updates to be synchronized once internet connectivity is restored.
- Facilitates guest communication and notifications, including booking confirmations, payment reminders, and check-in reminders.
- Supports multi-property management, allowing users to manage multiple resorts under a single account.
- Maintains audit trails and activity logs to track critical system actions for accountability and security.

2.3 User Classes and Characteristics

EaseInn supports multiple user roles with different responsibilities and interaction patterns. The system is designed for non-technical users by providing clear workflows and role-based interfaces.

1.Admin / Owner

- Acts as the primary system administrator and business owner.
- Has full access to all system functionalities.
- Responsibilities include:
  - Managing resorts and properties.
  - Configuring rooms, pricing, meal plans, and seasonal offers.
  - Managing users, roles, and permissions.
  - Viewing comprehensive analytics and financial reports.
- Interacts with the system mainly through the web portal.
- Typically has limited technical expertise and requires an intuitive interface.

2. Manager

- Responsible for day-to-day operational management of the resort.
- Has restricted administrative access based on assigned permissions.
- Responsibilities include:
  - Managing bookings, check-ins, and check-outs.
  - Assigning and monitoring staff tasks.
  - Handling guest requests and operational issues.
  - Viewing operational dashboards and reports.
- Interacts primarily through the web portal.
- Possesses moderate technical familiarity.

3. Staff

- Includes housekeeping staff, maintenance staff, and service staff.
- Access is limited to assigned tasks and task-related information only.
- Responsibilities include:
  - Viewing assigned tasks and deadlines.

    - Updating task status (e.g., In Progress, Completed).
    - Completing subtasks and checklists.
    - Uploading images or notes as task evidence.
- Interacts with the system via the mobile application.
- Requires a simple, task-focused interface due to minimal technical knowledge.

4. Guest

- External user with no direct administrative access to the system.
- Interacts indirectly with the system.
- Responsibilities and interactions include:
    - Receiving booking confirmations and invoices.
    - Making online payments via secure payment links.
    - Receiving notifications and reminders.
- May access a limited guest-facing portal (optional feature).
- Does not require technical knowledge.

2.4Operating Environment

EaseInn is designed for cloud deployment and access via web and mobile devices. The operating environment supports common SME hardware and typical network conditions in Sri Lanka.

Hardware Requirements

- Desktop or laptop computers for owners and managers.
- Smartphones or tablets for staff members.
- Optional printers for printing invoices and reports.

Software Requirements

- Web browsers: Google Chrome, Mozilla Firefox, Microsoft Edge (latest versions).
- Mobile operating systems:
  - Android (version 8.0 or higher).
  - iOS (version 13 or higher).
- Backend runtime environment:
  - Node.js (for backend services).
- Database systems:
  - MySQL for relational data.
  - MongoDB for unstructured data.

Network Requirements

- Stable internet connection for real-time system access.
- HTTPS-enabled communication for secure data transmission.
- Support for limited offline functionality in the staff mobile application with automatic synchronization when connectivity is restored.

Supported Platforms

- Cloud hosting platform: Amazon Web Services (AWS).
- Web-based access through modern browsers.
- Cross-platform mobile support via Flutter / React Native.

External Tools and Services

- Payment gateways (e.g., PayHere, Stripe) for online payments.
- Authentication and identity management service (Keycloak).
- Email and SMS notification services for guest and staff communications.
- Cloud storage services for document and image uploads.
- Containerization using Docker for deployment and scalability.

2.5 Design and Implementation Constraints

- The system must be completed within a one-year academic timeline as required by IS 3920.
- Development is limited by a restricted budget, requiring the use of open-source tools and free-tier services.
- The solution must use technologies suitable for a single-developer project.
- The system must comply with basic data protection and privacy practices.
- Dependence on third-party services such as payment gateways and notification providers may affect system functionality.
- The system must remain simple and user-friendly for non-technical users.
- Performance may be affected in low or unstable internet environments.

2.6 User Documentation

- A user manual for owners and managers explaining system setup, booking management, and reporting.
- A mobile application guide for staff covering task viewing, updates, and notifications.
- Quick-start guides for common tasks such as creating bookings and assigning tasks.
- Online help pages and FAQs accessible within the system.
- Basic training materials provided during system deployment.

2.7 Assumptions and Dependencies

Assumptions

- Owners/managers have access to a web-enabled device during operations.
- Staff members have smartphones available during work shifts.
- Users have basic digital literacy and can follow guided workflows.
- Room inventory and pricing are defined and maintained by the resort.

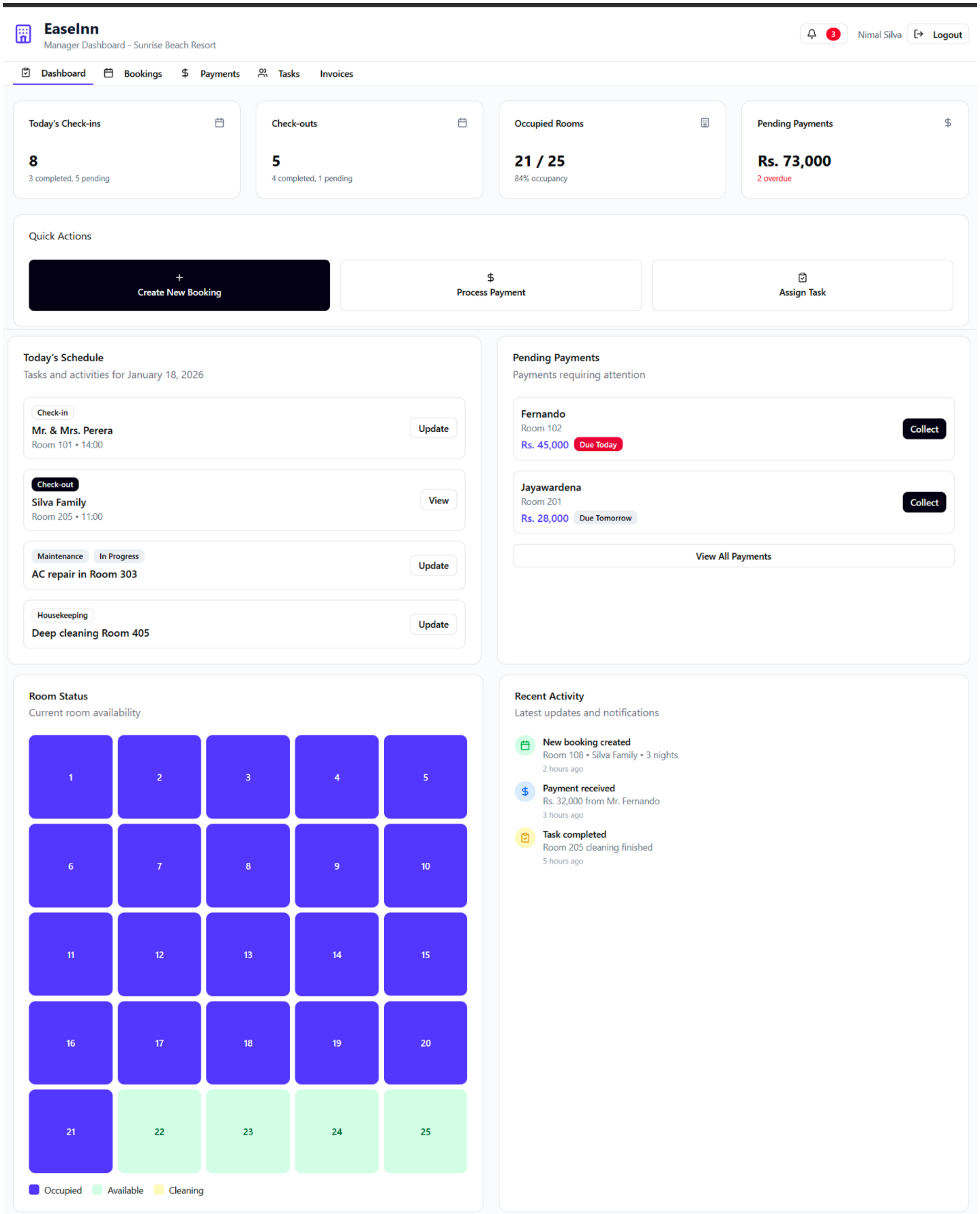
Dependencies

- Availability of AWS cloud infrastructure and database services.
- Payment gateway uptime and correct webhook/callback behavior.
- Email/SMS provider uptime and deliverability.
- Mobile OS support for notifications and background sync.
-

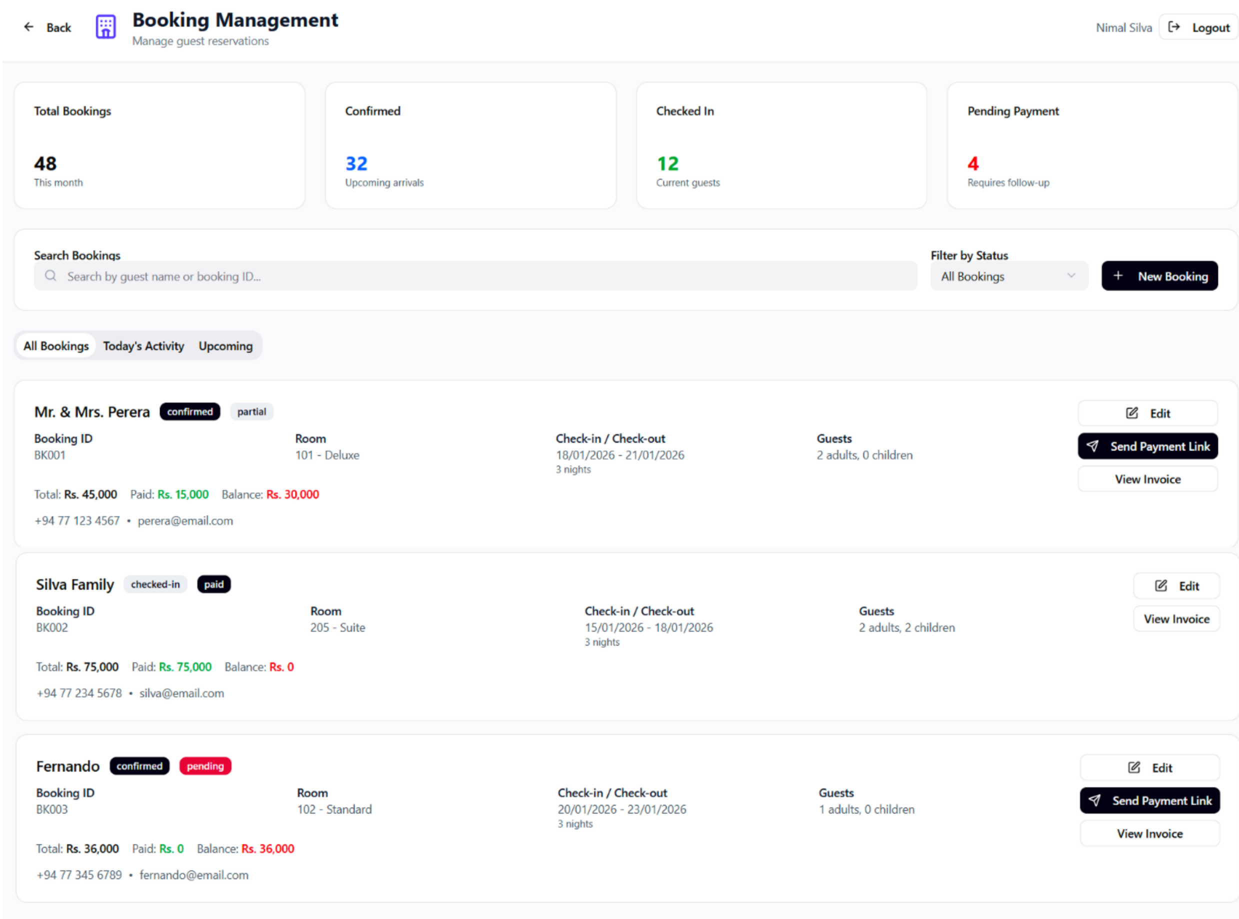
3 External Interface Requirements

3.1 User Interfaces

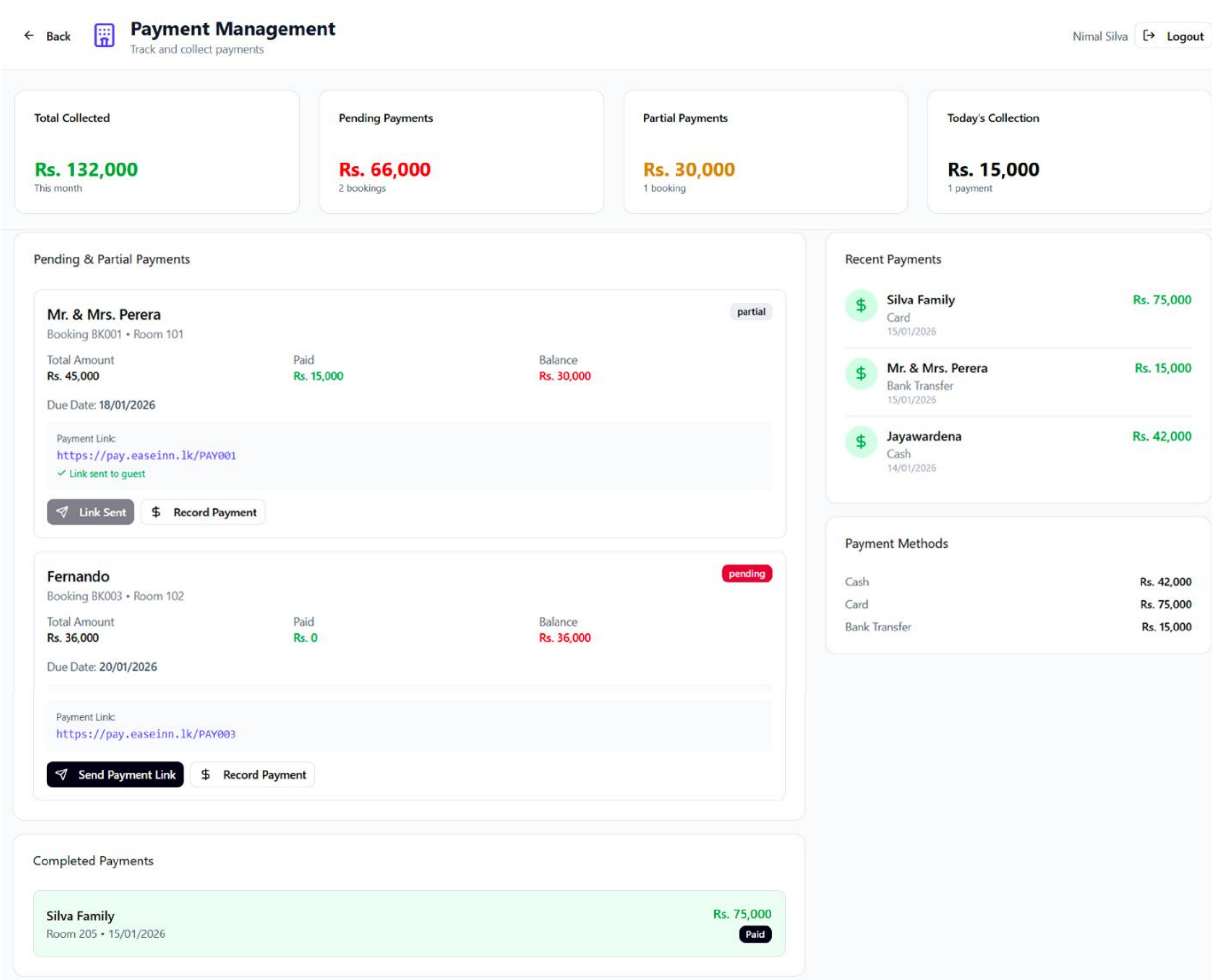
- The system provides a web-based user interface for owners and managers with a responsive and intuitive design.
- The web dashboard includes:
  - Calendar-based booking views.
  - Room availability and status indicators.
  - Dashboards for bookings, payments, and analytics.
- A mobile application interface is provided for staff with:
  - Task lists and status indicators.
  - Simple task update screens.
  - Push notification support.
- The user interfaces are designed to be:
  - Easy to use for non-technical users.
  - Consistent across devices.
  - Optimized for both desktop and mobile screens



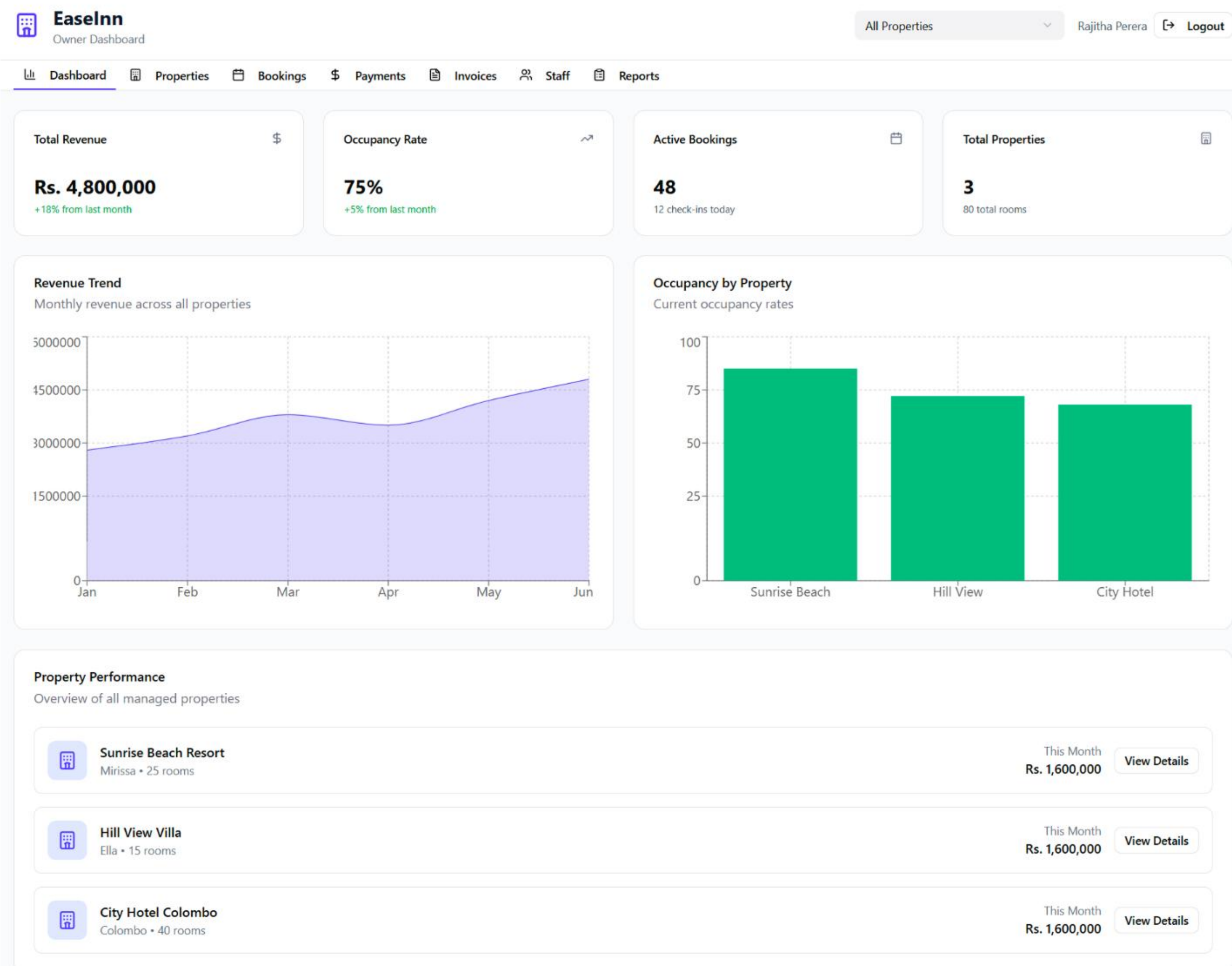
Manager Dashboard Interface



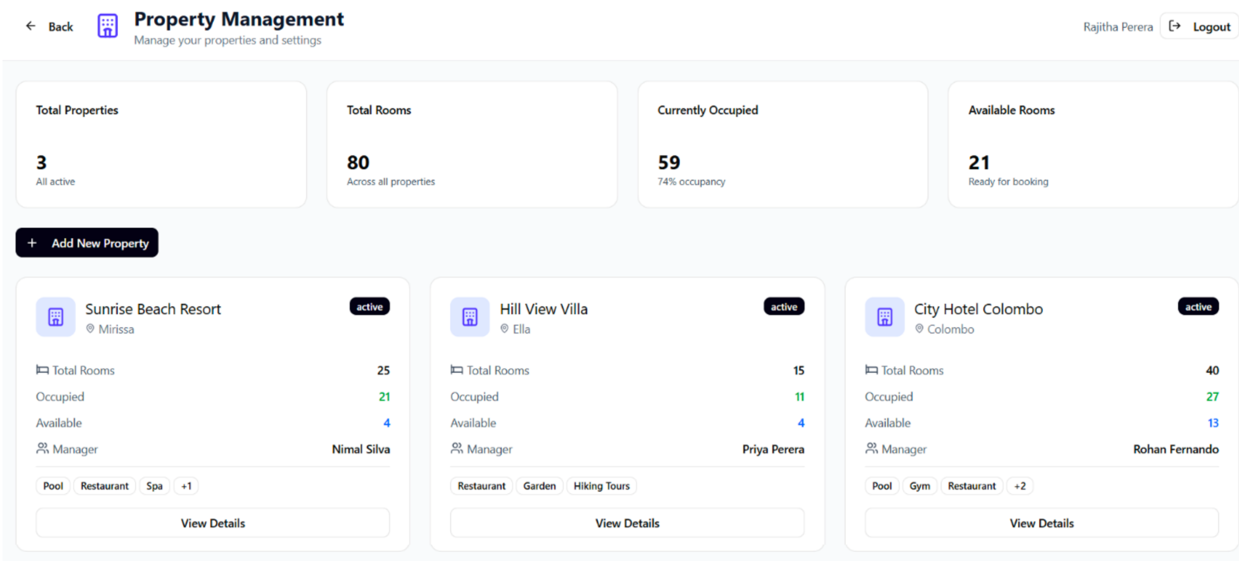
Booking Management Interface



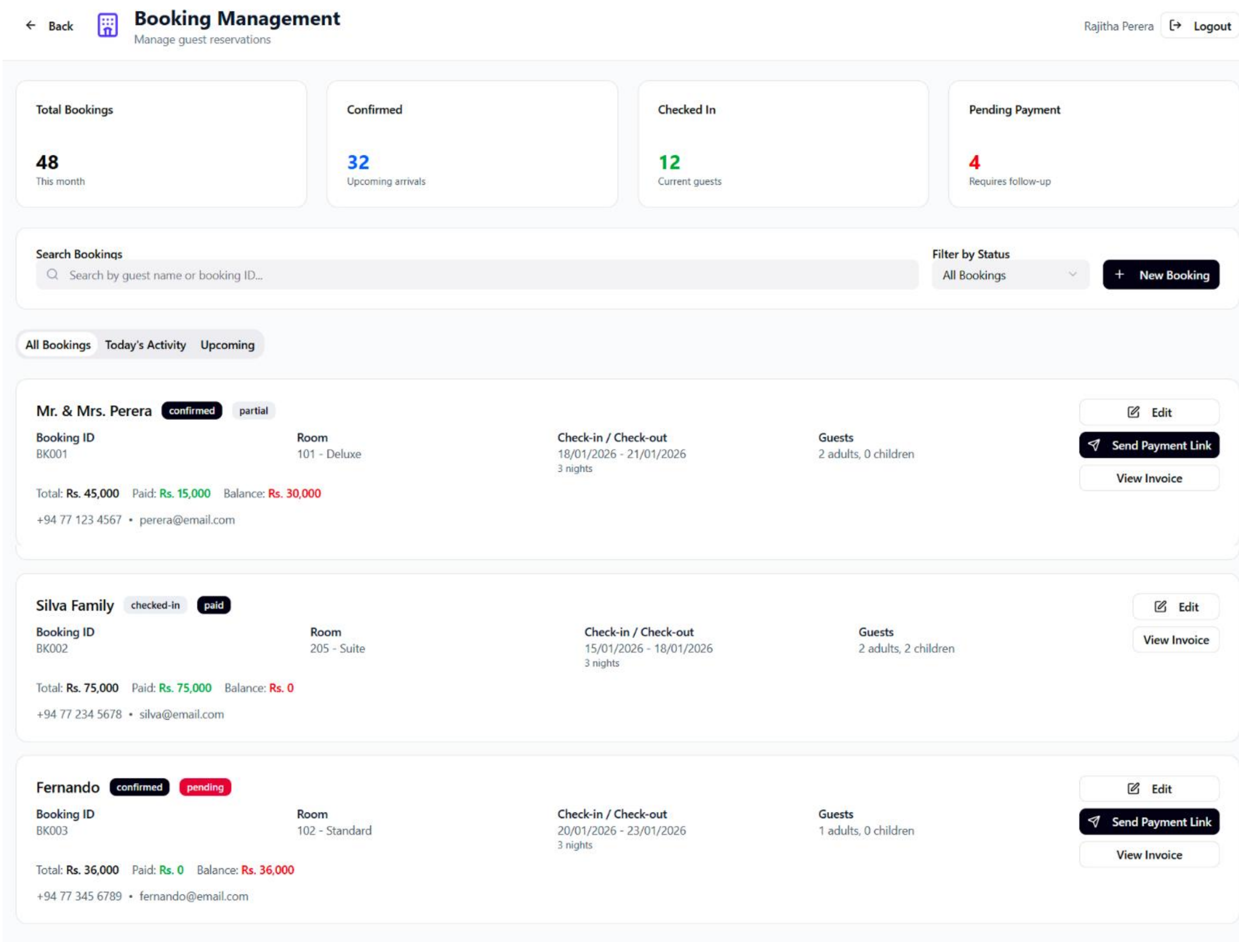
Payment Management Interface

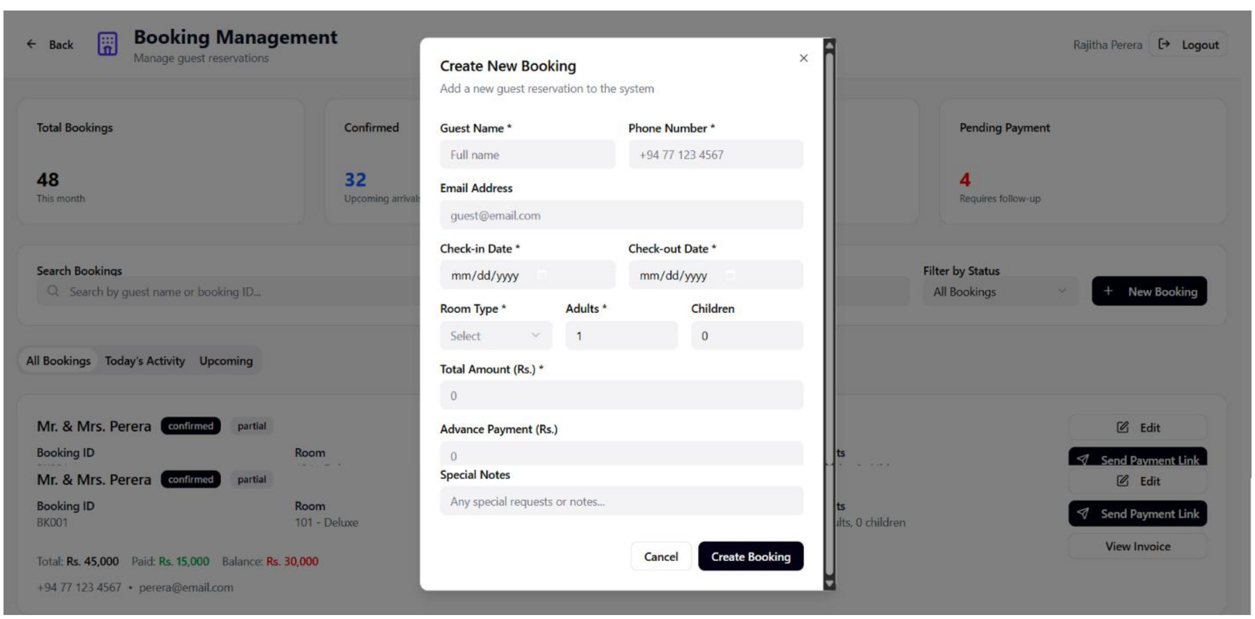


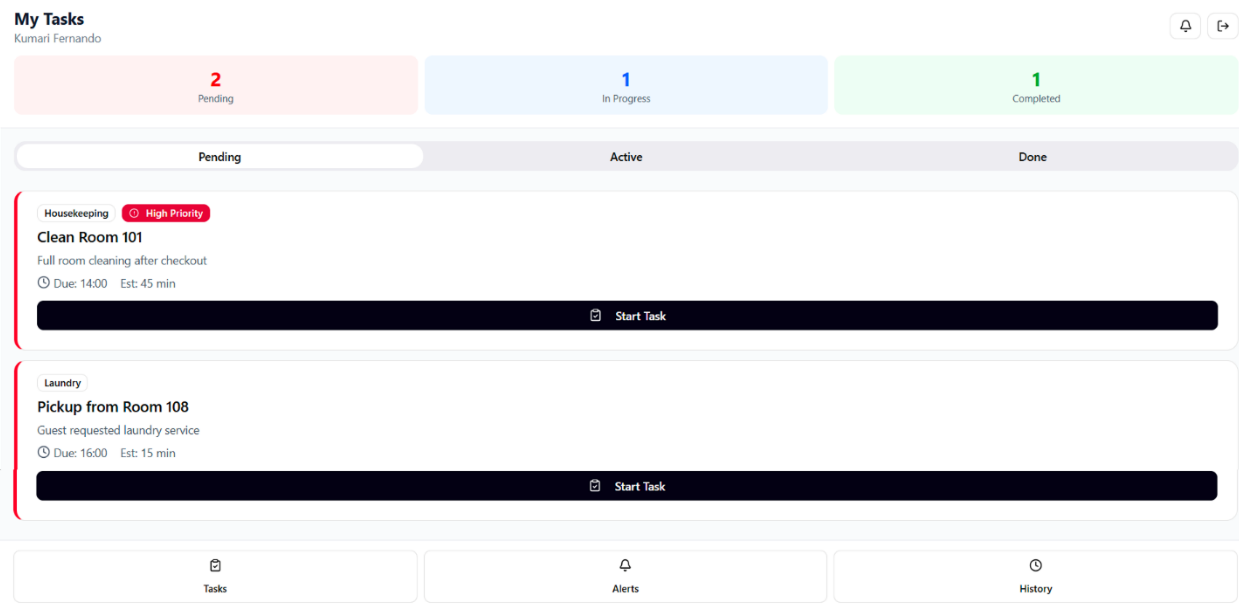
Owner Dashboard Interface



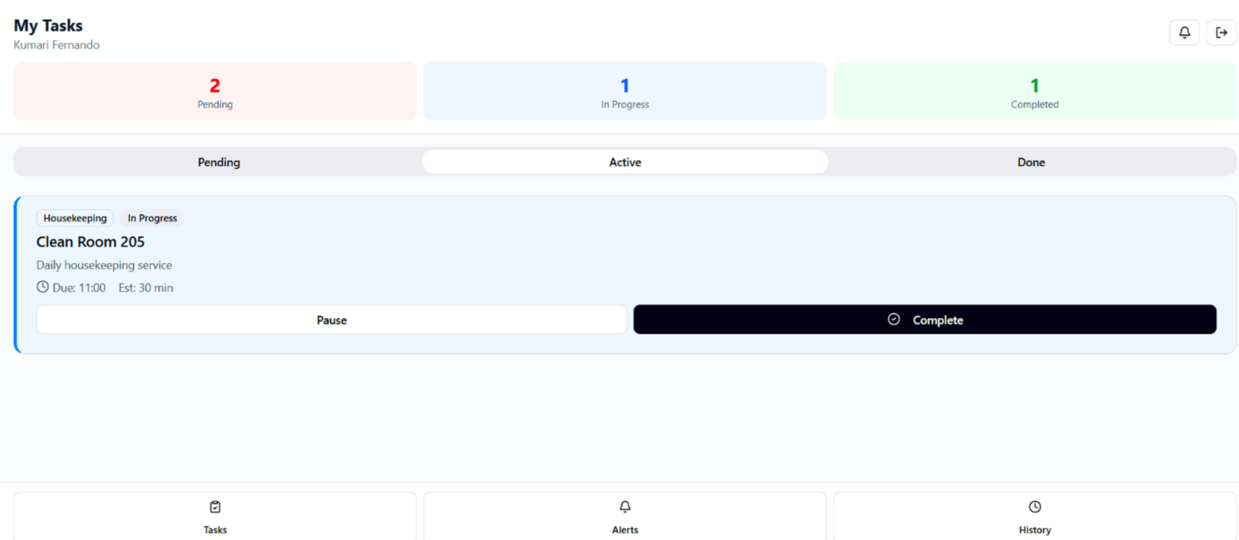
Property Management Interface


Back
Booking Management
Manage guest reservations
Rajitha Perera
Logout
Total Bookings
48
This month
Confirmed
32
Upcoming arrivals
Checked In
12
Current guests
Pending Payment
4
Requires follow-up
Search Bookings
Search by guest name or booking ID...
Filter by Status
All Bookings
New Booking
All Bookings
Today's Activity
Upcoming
Mr. & Mrs. Perera
confirmed
partial
Booking ID
BK001
Room
101 - Deluxe
Check-in / Check-out
18/01/2026 - 21/01/2026
3 nights
Guests
2 adults, 0 children
Edit
Send Payment Link
View Invoice
Total: Rs. 45,000 Paid: Rs. 15,000 Balance: Rs. 30,000
+94 77 123 4567 • perera@email.com
Silva Family
checked-in
paid
Booking ID
BK002
Room
205 - Suite
Check-in / Check-out
15/01/2026 - 18/01/2026
3 nights
Guests
2 adults, 2 children
Edit
View Invoice
Total: Rs. 75,000 Paid: Rs. 75,000 Balance: Rs. 0
+94 77 234 5678 • silva@email.com
Fernando
confirmed
pending
Booking ID
BK003
Room
102 - Standard
Check-in / Check-out
20/01/2026 - 23/01/2026
3 nights
Guests
1 adults, 0 children
Edit
Send Payment Link
View Invoice
Total: Rs. 36,000 Paid: Rs. 0 Balance: Rs. 36,000
+94 77 345 6789 • fernando@email.com

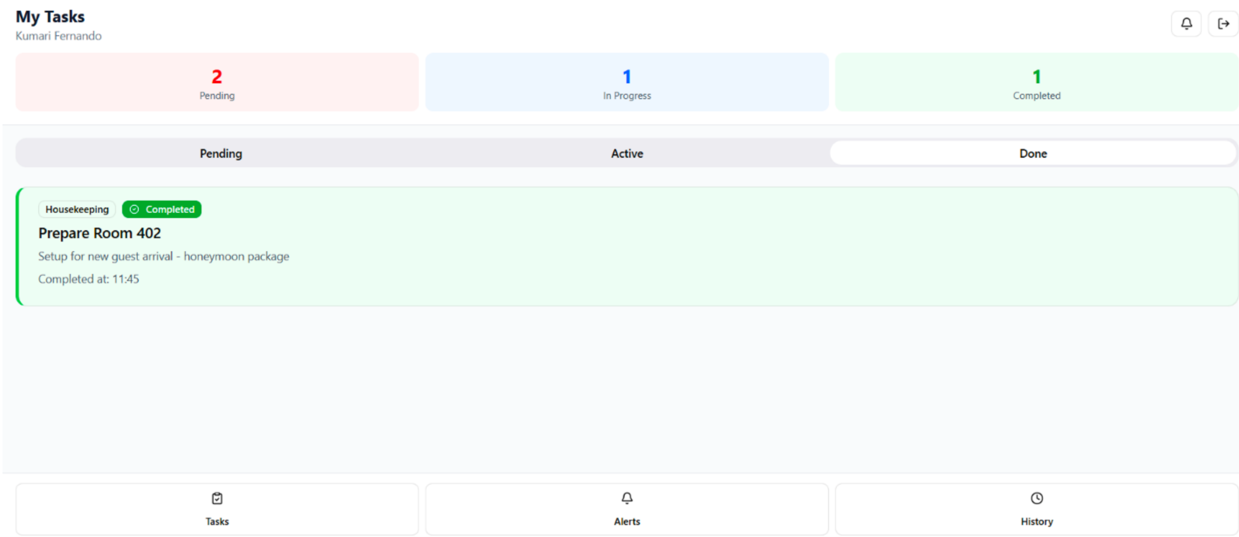

Create New Booking
Add a new guest reservation to the system
Guest Name *
Full name
Phone Number *
+94 77 123 4567
Email Address
guest@email.com
Check-in Date *
mm/dd/yyyy
Check-out Date *
mm/dd/yyyy
Room Type *
Select
Adults *
1
Children
0
Total Amount (Rs.) *
0
Advance Payment (Rs.)
0
Special Notes
Any special requests or notes...
Cancel
Create Booking



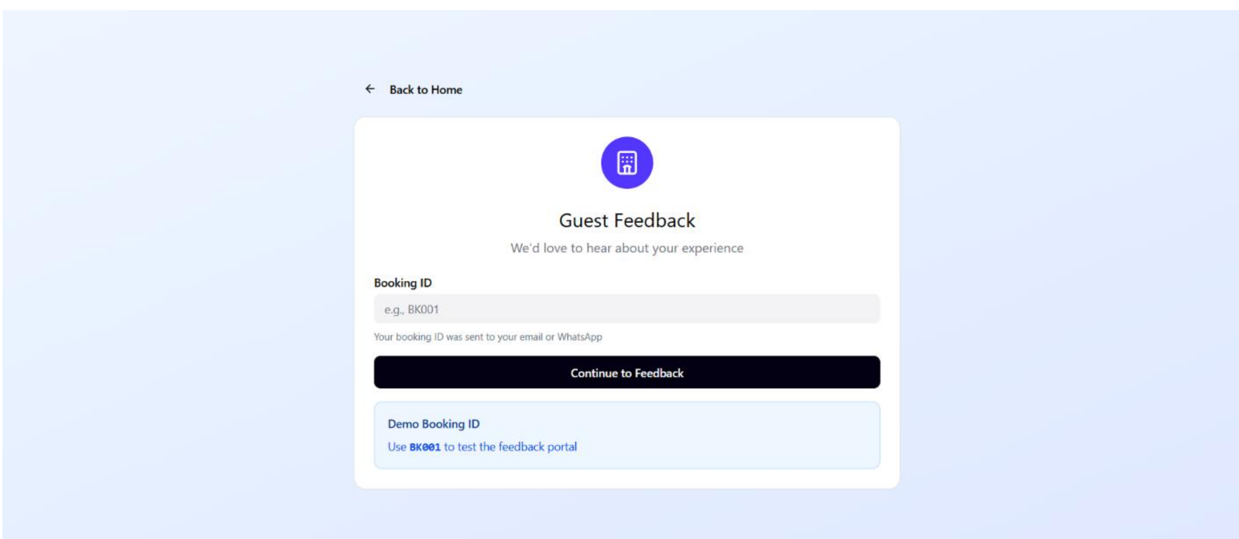
Task Management Interface



Task Management Interface



Task Management Interface



Feedback Interface

3.2 Hardware Interfaces

- Desktops/laptops for web portal access by owners and managers.
- Smartphones/tablets for staff mobile app usage.
- Mobile device camera for uploading guest ID documents and task evidence images.
- Optional printer for printing invoices and operational reports.

3.3 Software Interfaces

- Authentication: Keycloak for identity and access management.
- Payments: PayHere/Stripe APIs for payment link generation and confirmations.
- Notifications: Email SMTP/API and SMS gateway integration.
- Storage: object storage service for document and image uploads.
- Databases: MySQL and MongoDB as internal system components.

3.4 Communication Interfaces

- All data transmission between clients and servers is conducted using the HTTPS protocol to ensure secure communication.
- The system uses RESTful APIs with JSON as the standard data exchange format between the web application, mobile application, and backend services.
- Token-based authentication (e.g., JWT) is implemented to verify user identity and authorize API requests.
- Data is protected during transmission using TLS encryption, preventing unauthorized access and data interception.
- The system includes basic error handling mechanisms to manage network failures, API timeouts, and service unavailability, ensuring graceful failure and user-friendly error messages.

4 System Features

System features are described with description, priority, response sequence, and functional requirements. Requirements are written to be testable and aligned with project objectives.

4.1 Authentication, Authorization and Audit Logging

Description

This feature provides secure access to EaseInn. Users authenticate using credentials and are authorized based on roles such as Admin/Owner, Manager, and Staff. Role-based access control ensures each user sees only the screens and data relevant to their responsibilities. This reduces accidental changes and protects sensitive data.

The audit trail records critical activities such as booking creation, modification, cancellation, payment updates, pricing changes, and task completion. Audit logs support accountability and also help in investigating operational disputes (e.g., who modified a booking date).

Priority
High

Response Sequence

- User enters credentials in the web portal or mobile app.
- System validates credentials using the authentication service.
- System issues an access token and identifies user role and permissions.
- System redirects the user to the role-specific landing page.
- System logs the login event and subsequent critical actions.

Functional Requirements

- The system shall authenticate users using secure credential validation.
- The system shall enforce role-based access control for all protected operations and data views.
- The system shall allow Admin/Owner users to create, update, deactivate, and assign roles to user accounts.
- The system shall implement secure password policies and support password reset.
- The system shall enforce session timeout and token expiry for inactive sessions.
- The system shall log critical actions on bookings, payments, rooms, pricing, and tasks with timestamp and user ID.
- The system shall allow Admin/Owner users to view audit logs and filter by date range, user, and action type.
- The system shall lock or rate-limit repeated failed login attempts to reduce brute-force risk.

4.2 Property, Room and Pricing Management

Description

This feature allows owners and managers to configure properties and room inventory. Each room includes attributes such as type, capacity, amenities, and availability status. Rooms can be marked as unavailable due to maintenance, renovation, or other operational reasons.

Pricing management supports base prices, seasonal rates, meal plan surcharges, and promotional discounts. Accurate configuration is essential for correct invoicing and for reliable availability search.

Priority

High

Response Sequence

- Admin/Manager opens property settings and selects a property.
- User adds or edits rooms and room types with attributes and capacity.
- User defines pricing rules and seasonal rates for date ranges.
- System validates pricing rules and persists the configuration.
- System updates availability calculations and logs changes.

Functional Requirements

- The system shall allow Admin/Owner users to create and manage multiple properties under one account.
- The system shall allow users to add, update, deactivate, and categorize rooms by room type.
- The system shall allow rooms to be marked as Available, Booked, or In-Maintenance.
- The system shall allow configuration of base rates and seasonal rates by date range.
- The system shall support meal plan pricing and add-on services (optional items).
- The system shall support promotional discounts with start/end dates and eligibility rules.
- The system shall validate that pricing rules do not overlap in conflicting ways for the same room type.
- The system shall record configuration changes in the audit log.

4.3 Booking Management and Calendar Scheduling

Description

This feature supports the reservation lifecycle from initial availability search to booking confirmation, modification, cancellation, and checkout. A calendar-based interface allows managers to quickly understand occupancy, identify gaps, and allocate rooms efficiently.

Booking records include guest identity information, contact details, number of guests, selected room(s), dates, pricing breakdown, payment status, and operational notes. The system prevents double bookings by validating date overlaps and by respecting maintenance blocks.

Priority

High

Response Sequence

- User selects property, date range, and room type for availability search.
- System checks inventory, existing bookings, and maintenance blocks to compute availability.
- User selects room(s), enters guest details, and confirms the booking.
- System validates business rules and saves booking details.
- System sends booking confirmation and updates calendar occupancy.

Functional Requirements

- The system shall provide a calendar view of bookings per room and date.
- The system shall allow creation of bookings with guest details, dates, rooms, and pricing information.
- The system shall prevent double booking by validating overlapping date ranges for the same room.
- The system shall support booking modification with proper validation and an audit trail of changes.
- The system shall support booking cancellation with cancellation reason and timestamp.
- The system shall allow attaching guest documents (e.g., ID images) to bookings with secure access controls.
- The system shall support check-in and check-out status tracking for operational visibility.
- The system shall provide search and filters for bookings by guest name, date, status, and payment state.

4.4 Payments, Receipts and Invoicing

Description

This feature manages financial transactions for bookings. It supports generating payment requests, tracking advance payments and balances, and issuing invoices/receipts. Integration with a payment gateway enables guests to pay online without manual bank transfers, reducing delays and errors.

The system records gateway transaction references and reconciles payment confirmations via webhooks/callbacks. Invoices contain booking details, pricing breakdown, tax/service charges (if applicable), and payment history.

Priority

High

Response Sequence

- Manager opens booking and selects 'Request Payment'.
- System generates a payment link and stores gateway reference.
- Guest pays using the link and gateway processes the payment.
- Gateway sends confirmation callback to the system.
- System updates booking payment status and generates invoice/receipt.
- System sends confirmation and invoice to guest.

Functional Requirements

- The system shall allow recording advance payments and final payments for a booking.
- The system shall generate secure payment links via the configured payment gateway.
- The system shall store payment gateway transaction references and timestamps.
- The system shall validate payment webhook authenticity to prevent spoofing.
- The system shall automatically update payment status when payment confirmation is received.
- The system shall generate invoices and receipts with resort branding and a unique invoice number.
- The system shall allow managers to resend invoices and payment links to guests.
- The system shall provide a payment summary view showing paid amount, outstanding balance, and due date.

4.5 Staff and Task Management (Web and Mobile)

Description

This feature enables structured staff coordination for housekeeping, maintenance, and guest service tasks. Tasks can be linked to rooms or bookings so that staff understand context and deadlines (e.g., clean Room 05 before 2:00 PM check-in).

The mobile app allows staff to view assigned tasks, update progress, complete checklists, and upload evidence. Offline-first behavior ensures updates are not lost during connectivity issues.

Priority

High

Response Sequence

- Manager creates a task with priority, deadline, and optional room/booking link.
- Manager assigns the task to a staff member.
- System sends a notification to the staff member.
- Staff opens task detail, completes checklist items, and updates status.
- System synchronizes updates to web dashboard and records completion timestamp.
- Manager reviews completion and closes the task.

Functional Requirements

- The system shall allow managers to create tasks with title, description, priority, deadline, and assignee.
- The system shall support task status values such as Open, In Progress, Blocked, and Completed.
- The system shall support subtasks/checklists and prevent task closure until checklist completion (configurable).
- The system shall allow linking tasks to rooms and bookings.
- The system shall notify staff of assigned tasks and urgent updates via push notifications.
- The system shall allow staff to add notes and upload images as evidence.
- The system shall support offline task updates and synchronize when connectivity is restored.
- The system shall provide a manager view of overdue tasks and escalation alerts (e.g., highlight overdue tasks).

4.6 Analytics, Reports and AI Insights

Description

This feature provides dashboards and reports to support decision-making. Owners can review occupancy and revenue trends, identify peak periods, and evaluate performance over time. Operational analytics help managers plan staffing and maintenance. Reports can be filtered by property and room type.

AI insights are optional and provide advisory suggestions such as expected demand and recommended pricing adjustments based on historical bookings. These suggestions are meant to assist managers and do not automatically change prices.

Priority

Medium

Response Sequence

- Owner selects reporting period and property filters.

- System aggregates bookings, payments, and occupancy metrics.
- System displays KPIs and charts/tables on dashboard.
- User exports report or downloads summary.
- System displays optional AI insights based on historical patterns.

2.3.1 Functional Requirements

- The system shall display occupancy rate and available room count for selected dates.
- The system shall generate revenue summaries for daily, weekly, and monthly periods.
- The system shall provide booking trend reports including cancellations and peak seasons.
- The system shall provide staff performance metrics based on tasks assigned and completed.
- The system shall support exporting reports to CSV/PDF.
- The system should provide AI-based insights such as demand forecasting and pricing suggestions.

4.7 Guest Communication and Feedback Management

Description

This feature ensures consistent communication with guests through confirmations, reminders, and invoices. Clear communication reduces missed check-ins and improves guest satisfaction. The system can send messages through email and SMS using configurable templates.

After checkout, the system may request feedback. Feedback is stored and reviewed by managers to improve service quality. In future, feedback can be categorized or summarized using AI.

Priority

Medium

Response Sequence

- System sends booking confirmation after booking creation.
- System sends payment reminder if payment is pending.
- System sends check-in reminder prior to arrival.
- After checkout, system sends feedback request link.
- Guest submits feedback and system notifies manager.

Functional Requirements

- The system shall send booking confirmations and invoices to guests via email/SMS.
- The system shall send payment reminders based on configurable rules.
- The system shall allow managers to configure message templates and branding elements.

- The system shall allow guests to submit feedback via a secure link without requiring an account.
- The system shall store feedback linked to a booking and allow managers to view it in a dashboard.

4.8 Room Maintenance and Housekeeping Scheduling

Description

This feature supports operational control of room readiness by allowing managers to mark rooms as under maintenance, schedule housekeeping windows, and track maintenance requests. Maintenance blocks should be reflected in availability calculations so that rooms are not accidentally booked when unavailable.

Housekeeping scheduling can be derived from upcoming check-ins/check-outs and can automatically suggest tasks such as cleaning, linen changes, and inspection. This improves operational efficiency and ensures rooms are prepared on time.

Priority

Medium

Response Sequence

- Manager records a maintenance issue for a room and sets the room status to In-Maintenance.
- System blocks the room from availability search for the specified dates.
- Manager assigns a maintenance task to staff and monitors progress.
- After completion, manager marks the room as Available and logs the maintenance record.
- System updates availability and includes maintenance history in reports.

Functional Requirements

- The system shall allow managers to mark rooms as In-Maintenance with a date range and reason.
- The system shall exclude rooms marked as In-Maintenance from availability search.
- The system shall allow managers to create maintenance requests and assign them as tasks.
- The system shall retain a maintenance history for each room.

4.9 Multi-Property and Centralized Management

Description

This feature supports owners managing multiple resorts or accommodation units. It allows switching between properties, applying consistent policies, and viewing consolidated analytics. Multi-property support is important for scalability and business growth.

The system should support centralized user management where staff accounts can be assigned to one or more properties, with permissions scoped appropriately. Consolidated reporting provides a single view of performance across locations.

Priority

Low

Response Sequence

- Owner creates a second property under the account.
- Owner configures rooms and pricing for the new property.
- Owner assigns managers and staff to the property.
- System provides property switch controls and consolidated reporting.
- Owner generates cross-property reports for comparison.

Functional Requirements

- The system shall allow an owner account to manage more than one property.
- The system shall support assigning users to specific properties with scoped permissions.
- The system shall allow switching between properties in the web portal.
- The system shall provide consolidated occupancy and revenue reports across properties.
- The system shall ensure that property data is isolated and cannot be accessed by unauthorized users.

5 Other Nonfunctional Requirements

5.1 Performance Requirements

Performance requirements ensure that EaseInn remains responsive during peak operational periods, such as weekends and holiday seasons. Since reservations and task coordination are time-sensitive, delays can directly impact guest satisfaction and operational efficiency.

- The system shall load the main dashboard within 3 seconds under normal conditions.
- The system shall support at least 50 concurrent active sessions for SME-scale usage.
- The system shall process booking creation and validation within 5 seconds.

- The system shall process payment status update within 5 seconds after receiving a valid webhook callback.
- The system shall synchronize mobile task updates within 10 seconds for typical payload sizes when online.
- The system shall maintain uptime of at least 99% excluding scheduled maintenance.
- The system shall store uploaded documents efficiently and should limit file sizes to protect performance.

5.2 Safety Requirements

Safety requirements in EaseInn refer to data safety and operational safety. The system should avoid inconsistent states, prevent accidental destructive actions, and ensure that critical records such as bookings and payments are not lost.

- The system shall perform automated daily backups of databases and file storage references.
- The system shall provide confirmation dialogs for cancellation and deletion actions.
- The system shall ensure atomic booking creation so that partial failures do not create inconsistent bookings.
- The system shall provide rollback or compensating actions when payment confirmation fails after a guest pays.
- The system shall display clear error messages and recommended recovery steps.
- The mobile app shall queue offline updates and synchronize without overwriting newer server updates (conflict handling).

5.3 Security Requirements

EaseInn must protect guest personal information, payment references, and operational records. Security controls cover authentication, authorization, encryption, auditability, and secure storage.

- The system shall enforce strong authentication and password policies.
- The system shall use role-based authorization for every protected API and UI module.
- The system shall transmit data using TLS encryption over HTTPS.
- The system shall store sensitive references securely and restrict document access by role and booking context.
- The system shall log authentication attempts and suspicious activities.
- The system shall rate-limit repeated failed login attempts.
- The system shall verify payment gateway webhooks and reject unverified callbacks.
- The system should support optional two-factor authentication for Admin/Owner accounts.

5.4 Software Quality Attributes

EaseInn is designed to meet the following measurable quality attributes:

- Usability: users should complete common tasks (create booking, assign task) with minimal steps and without specialized training.
- Reliability: the system should prevent double bookings and ensure accurate payment tracking under normal use.
- Maintainability: modular architecture and clean APIs should allow new features to be added with minimal impact.
- Scalability: the system should support additional properties and increased booking volume without major redesign.
- Portability: web portal should work across major browsers; mobile app should support both Android and iOS.
- Availability: the platform should be accessible during business hours with minimal downtime.

Quality assurance will be supported through unit testing of key business rules, integration testing of payment and notification interfaces, and usability testing with representative users.

5.5 Business Rules

Business rules enforce correct operational behavior and protect the resort from common mistakes:

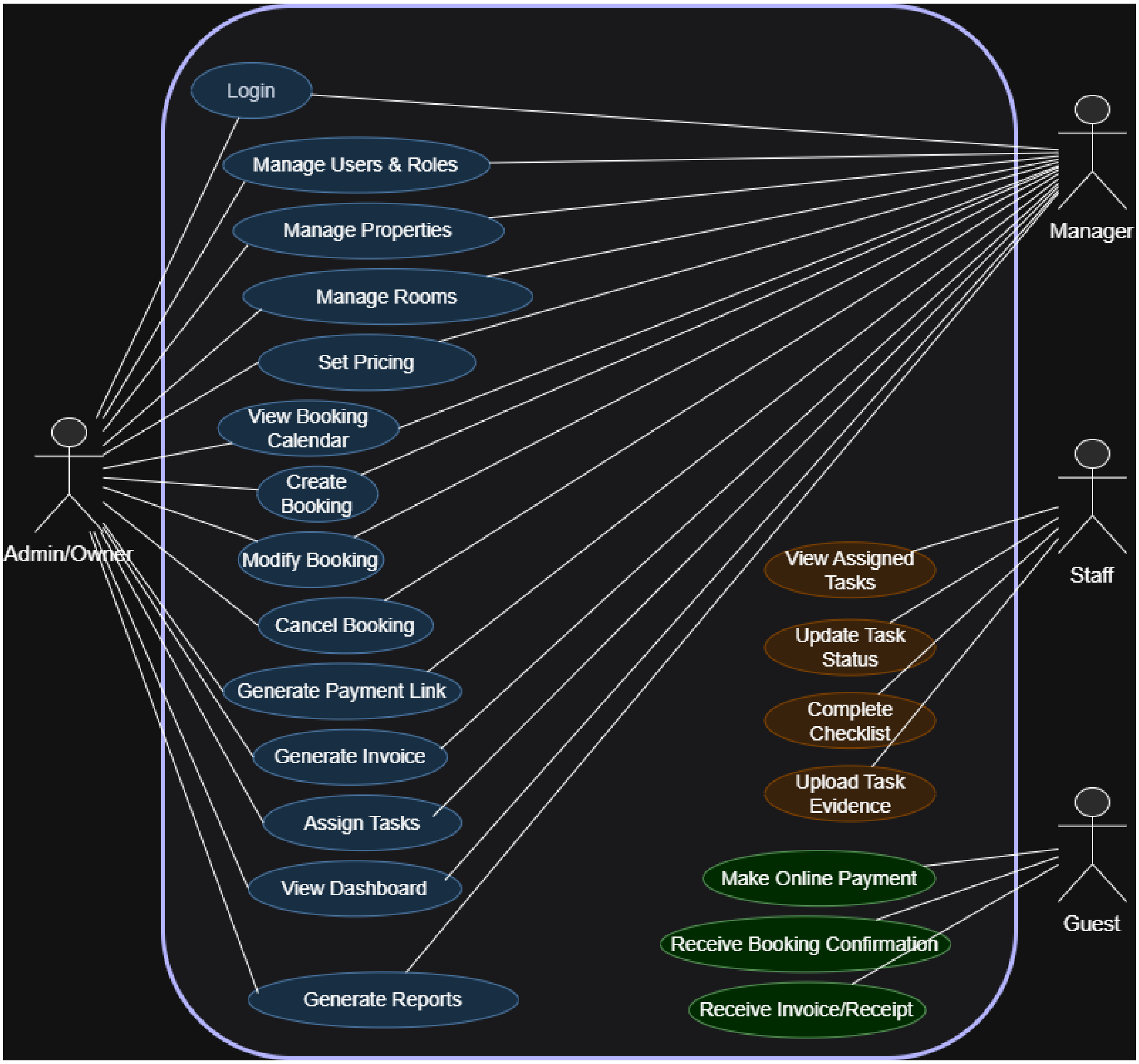
- A room shall not be assigned to more than one confirmed booking for overlapping dates.
- A room marked as In-Maintenance shall not be available for booking.
- A booking shall not be marked as fully paid unless total recorded payments equal or exceed the total charge.
- Only authorized roles may modify pricing, room inventory, and cancellation policies.
- Tasks linked to an upcoming check-in should be completed before the check-in time where applicable.
- Canceled bookings shall be retained for reporting and audit purposes.
- Refund handling, if implemented, shall require managerial approval and follow the resort's policy.

Appendix A – Glossary

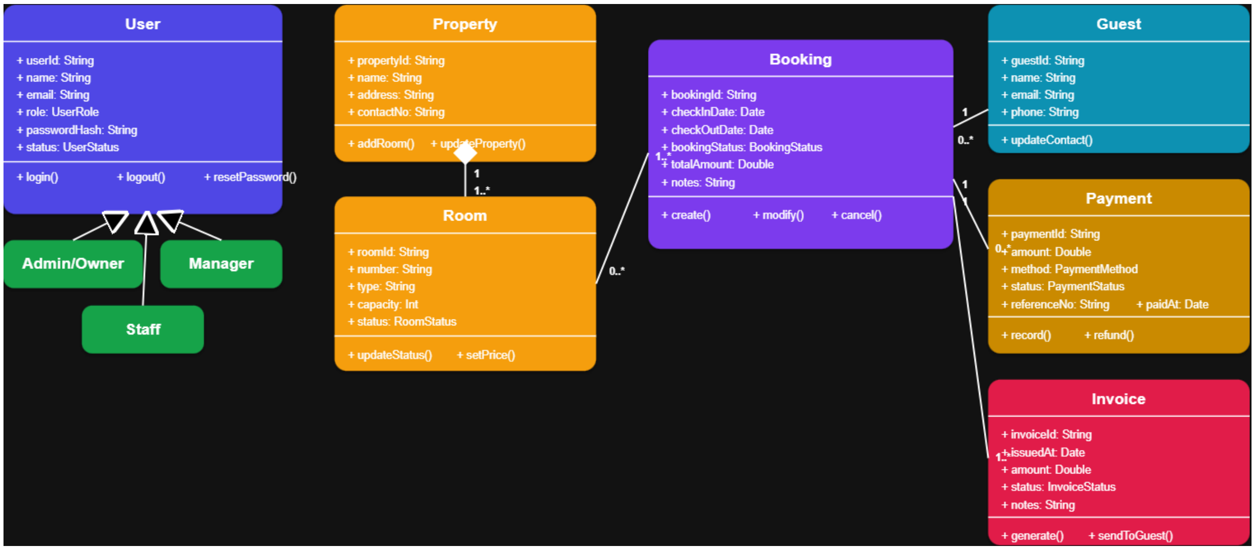
- EaseInn – The proposed resort booking and management system developed in this project.
- PMS (Property Management System) – Software used to manage hotel or resort operations such as bookings, rooms, and payments.
- SME (Small and Medium Enterprise) – Small to medium-scale business entities.
- Booking – A reservation made by a guest for a specific room and date range.
- Room Availability – The status indicating whether a room is available, booked, or under maintenance.
- Task – An operational activity assigned to staff, such as cleaning or maintenance.
- Subtask – A smaller task that forms part of a main task.
- API (Application Programming Interface) – A set of rules that allows software components to communicate.
- REST API – An API architecture using HTTP methods for communication.
- JSON – A lightweight data format used for data exchange.
- Authentication – The process of verifying user identity.
- Authorization – The process of granting access rights to users.
- Payment Gateway – A service that processes online payments securely.
- Invoice – A document generated to record payment details for a booking.
- Audit Trail – A record of system activities for security and accountability.

Appendix B – Analysis Models

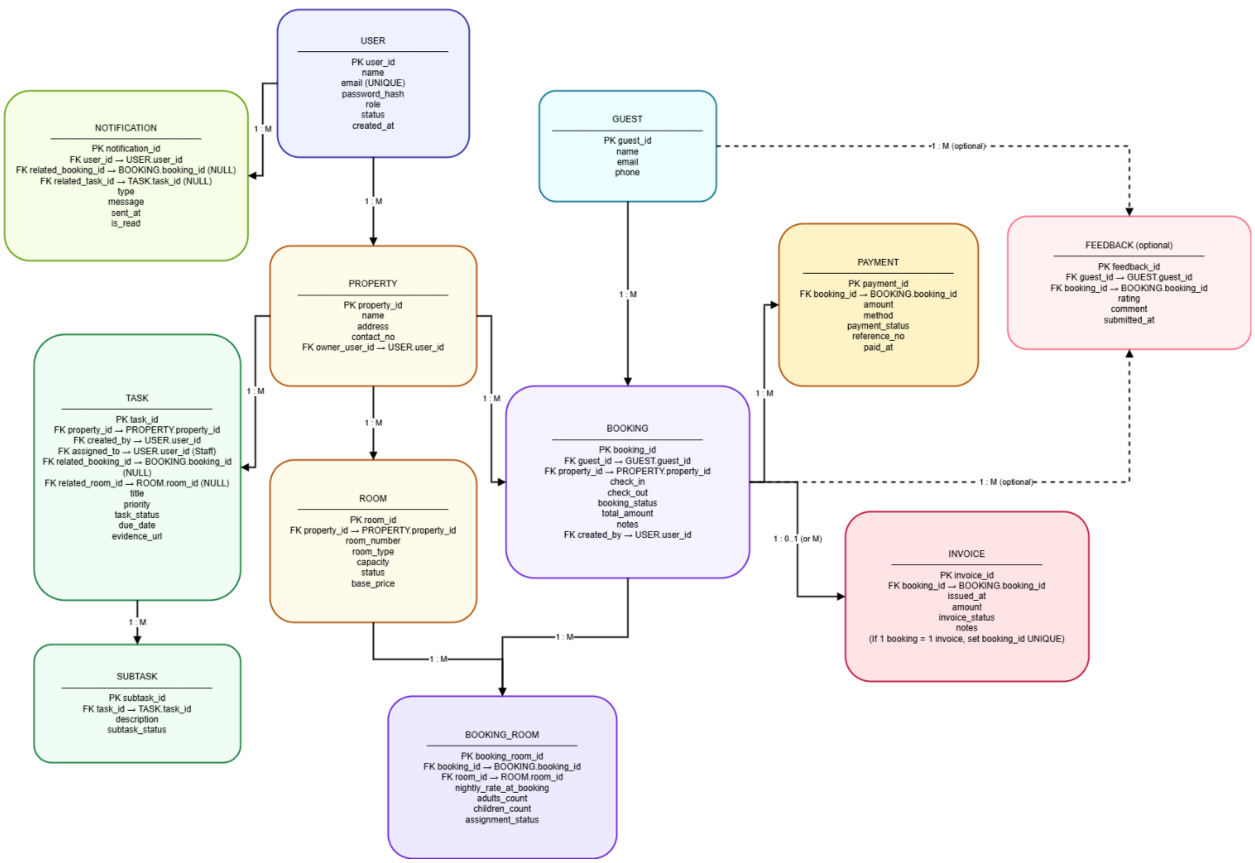
1 Use Case Diagram



2 Class Diagram



3 ER Diagram



Appendix C – Detailed Functional Requirement Catalogue

This appendix provides additional functional requirements that refine system behavior for completeness and testability. These requirements complement the feature-level requirements in Section 4 and are intended to support test case design.

Validation Rules and Error Handling

Validation rules prevent incorrect data entry and ensure consistent operations. The system shall provide clear messages and safe recovery paths when errors occur.

- The system shall validate that booking start date is earlier than booking end date.
- The system shall validate that the number of guests does not exceed the room capacity unless explicitly overridden by a manager.
- The system shall prevent saving a booking without a guest contact number or email.
- The system shall prevent marking a booking as checked-in if required documents are missing (configurable policy).
- The system shall prevent generating an invoice if pricing rules are incomplete for the booking date range.
- The system shall handle payment webhook duplicates idempotently to avoid double-recording payments.
- The system shall provide user-friendly error messages and shall log technical details for debugging.

Appendix D – Verification and Acceptance Overview

This appendix outlines how the requirements in this SRS will be verified during implementation. The goal is to ensure that each major feature can be demonstrated during evaluations and that the system behaves reliably under realistic usage conditions.

Verification Approach

Verification will be performed through a combination of unit testing, integration testing, and scenario-based demonstrations. Unit tests will validate core business rules such as preventing double bookings and correct pricing calculations. Integration tests will validate external interfaces such as payment gateway callbacks and notification sending. Scenario-based demonstrations will be used to show complete user journeys such as creating a booking, requesting an advance payment, assigning housekeeping tasks, and generating reports.

Acceptance Criteria (Examples)

The following acceptance criteria summarize key expected outcomes for a successful project delivery:

- A manager can create a booking and the system prevents assigning a room that is already booked for overlapping dates.
- A payment link can be generated and, upon successful payment, the booking status updates and an invoice can be produced.
- A manager can assign a housekeeping task to a staff member and the staff member can update the task using the mobile app (including offline sync).
- An owner can view occupancy and revenue reports for a selected period and export a report for accounting purposes.
- Audit logs show the user and timestamp for critical actions such as booking modification and pricing changes.

Appendix E – Future Enhancements

The initial EaseInn release focuses on core resort operations that provide the highest value within the academic timeframe. However, the architecture is designed to accommodate future enhancements. Potential improvements include channel integration with popular booking platforms, a guest self-service portal for viewing reservations and requesting services, and automated pricing optimization based on demand. Additional features may include inventory management for food and supplies, staff shift scheduling, and deeper analytics such as cohort analysis of guest types and repeat visit rates. From a technical perspective, enhancements may include stronger security features such as optional two-factor authentication, advanced monitoring and alerting, and improved offline conflict resolution for mobile synchronization. These enhancements are considered out of scope for the current academic project but can be implemented incrementally in future versions.